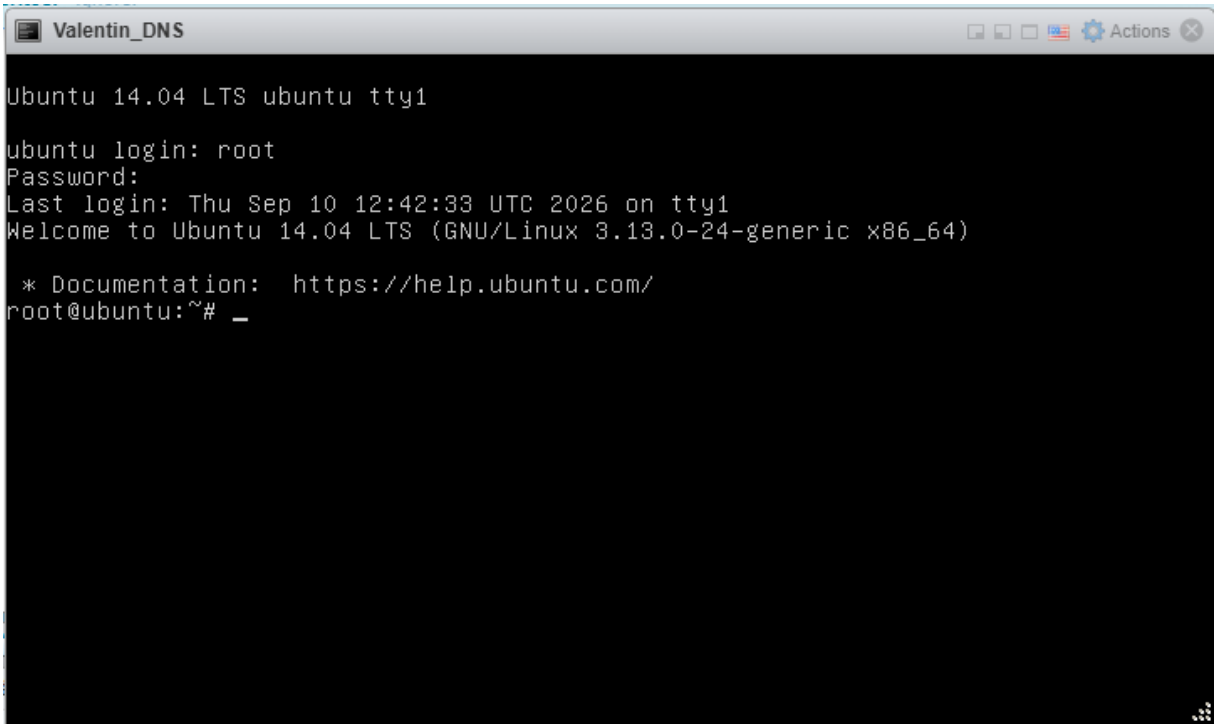
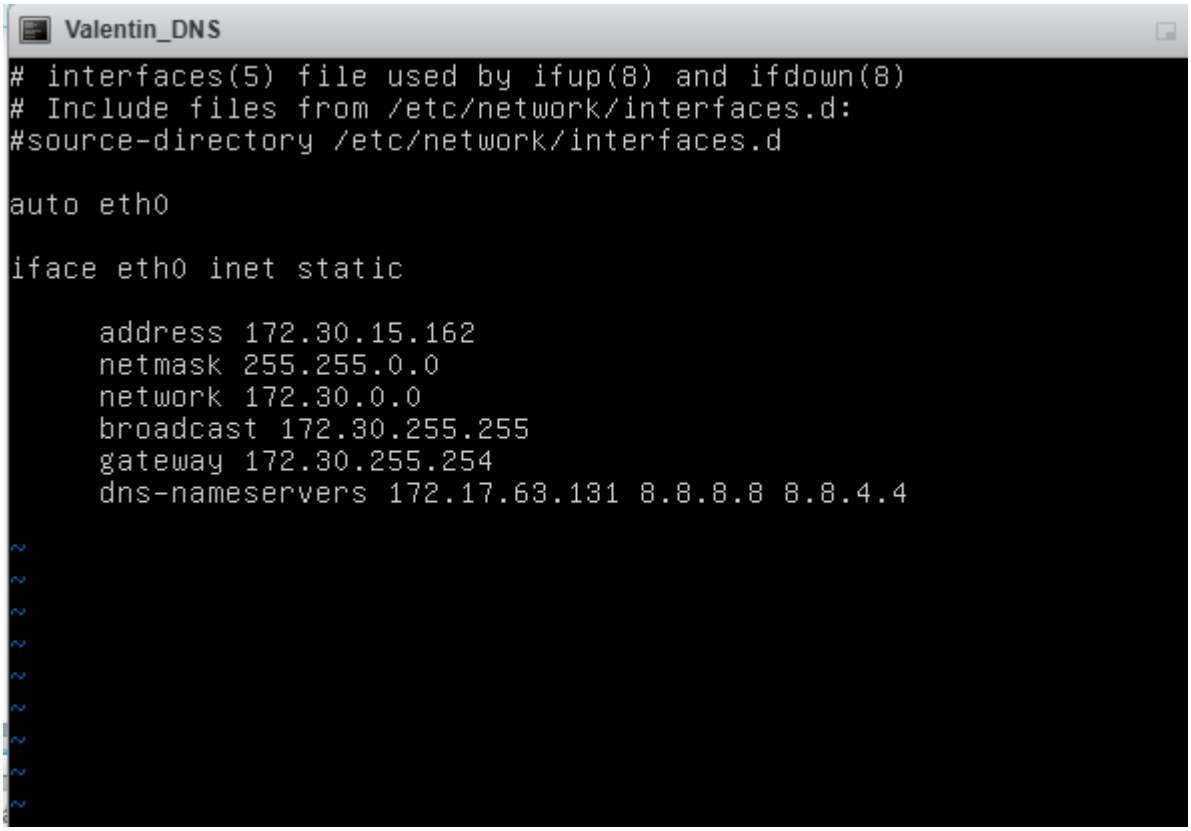
TP08 B2 Mise en place d'un serveur DNS :

Valentin Chaussegros.

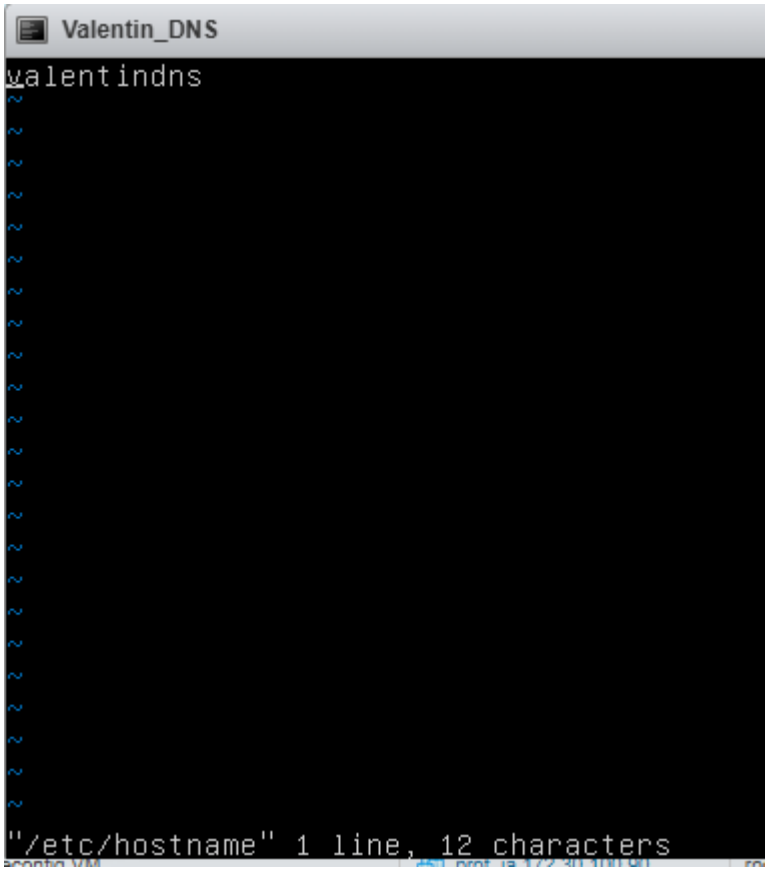
1) installation du serveur DNS sur une machine Ubuntu light :



On configure l'adresse IP de notre serveur :



On change le nom d'hôte :



Mise à jour et téléchargement des packages :

Après le redémarrage, on met à jour le cache des paquets puis l'ensemble du système :

```
Terminal — mise à jour du système

root@ubuntu:~# apt-get update
root@ubuntu:~# apt-get upgrade
```

On installe ensuite l'éditeur de texte **nano** :

```
Terminal — éditeur nano

root@ubuntu:~# apt-get install nano
```

Puis les utilitaires réseau pour le DNS, qui contiennent notamment `nslookup` :

```
Terminal — dnsutils

root@ubuntu:~# apt-get install dnsutils
```

Enfin, on installe le serveur DNS **Bind9** :

```
Terminal — bind9

root@ubuntu:~# apt-get install bind9
```

Configuration de Bind9 :

```
root@valentindns:~# cd /etc/bind
root@valentindns:/etc/bind# ls
bind.keys  db.empty    named.conf.default-zones  zones.rfc1918
db.0       db.local    named.conf.local
db.127     db.root     named.conf.options
db.255     named.conf  rndc.key
root@valentindns:/etc/bind#
```

On crée un dossier de sauvegarde, puis on y copie le fichier `named.conf.local` en cas de problème :

```
Terminal — sauvegarde

root@prof-dns:/etc/bind# mkdir sauv_fic_conf
root@prof-dns:/etc/bind# cp named.conf.local /etc/bind/sauv_fic_conf/
```

```
// This is the primary configuration file for the BIND DNS server named.
//
// Please read /usr/share/doc/bind9/README.Debian.gz for information on the
// structure of BIND configuration files in Debian, *BEFORE* you customize
// this configuration file.
//
// If you are just adding zones, please do that in /etc/bind/named.conf.local

include "/etc/bind/named.conf.options";
include "/etc/bind/named.conf.local";
include "/etc/bind/named.conf.default-zones";
```

Zone de résolution direct :

```
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     valentindns.btssio.com. root.valentindns.btssio.com. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@               IN      NS      valentindns.btssio.com.
valentindns     IN      A       172.30.15.162
client          IN      A       172.30.15.200
www             IN      CNAME   valentindns.btssio.com.
```

Zone de résolution inverse :

```
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     valentindns.btssio.com. root.valentindns.btssio.com. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@           IN      NS      valentindns.btssio.com.
162         IN      PTR     valentindns.btssio.com.
200         IN      PTR     client.btssio.com.
www         IN      CNAME   valentindns.btssio.com.
```

Déclaration des zones :

```
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "btssio.com" IN {
    type master;
    file "/etc/bind/btssio.com.zone";
};

zone "15.30.172.in-addr.arpa." IN {
    type master;
    file "/etc/bind/btssio.com.rev";
};
```

Configuration du client DNS :

```
# Dynamic resolv.conf(5) file for glibc resolver(3) generated by resolvconf(8)
#     DO NOT EDIT THIS FILE BY HAND -- YOUR CHANGES WILL BE OVERWRITTEN
nameserver 172.17.63.131
nameserver 8.8.8.8
nameserver 8.8.4.4
# Dynamic resolv.conf(5) file for glibc resolver(3) generated by resolvconf(8)
#     DO NOT EDIT THIS FILE BY HAND -- YOUR CHANGES WILL BE OVERWRITTEN
#nameserver 212.54.44.54
#nameserver 212.54.40.25
nameserver 172.30.15.162
search     btssio.com
```

Installation resolvconf :

```
root@valentindns: /etc/bind
  GNU nano 2.2.6      File: /

# Serveur DNS Principal
nameserver 172.30.15.162
search     btssio.com
```

Redémarrage bind9 :

Il vaut mieux redémarrer le service `bind9` afin de conserver et d'appliquer la configuration :

Terminal – redémarrage de bind9

```
root@prof-dns:/etc/bind# /etc/init.d/bind9 restart
 * Stopping domain name service... bind9
   waiting for pid 690 to die
 * Starting domain name service... bind9
root@prof-dns:/etc/bind# _
```

2) Test :

Test client – serveur :

```
C:\Users\CHAUSSEGROS Valentin>nslookup valentindns.btssio.com 172.30.15.162
Serveur :   valentindns.btssio.com
Address:  172.30.15.162

Nom :    valentindns.btssio.com
Address:  172.30.15.162


C:\Users\CHAUSSEGROS Valentin>nslookup client.btssio.com 172.30.15.162
Serveur :   valentindns.btssio.com
Address:  172.30.15.162

Nom :    client.btssio.com
Address:  172.30.15.200


C:\Users\CHAUSSEGROS Valentin>
```

Test CNAME :

```
C:\Users\CHAUSSEGROS Valentin>nslookup www.btssio.com 172.30.15.162
Serveur :   valentindns.btssio.com
Address:  172.30.15.162

Nom :    valentindns.btssio.com
Address:  172.30.15.162
Aliases:  www.btssio.com
```

Test nom inexistant :

```
C:\Users\CHAUSSEGROS Valentin>nslookup inconnu.btssio.com 172.30.15.162
Serveur :   valentindns.btssio.com
Address:  172.30.15.162

*** valentindns.btssio.com ne parvient pas à trouver inconnu.btssio.com : Non-existent domain

C:\Users\CHAUSSEGROS Valentin>
```